

Power Plants and PM_{2.5}

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1 Methods

Q1: Plants per county. 12,175 of 12,185 plants matched a county, meaning we were left with 10 plants that did not match a county.

Plants per U.S. County

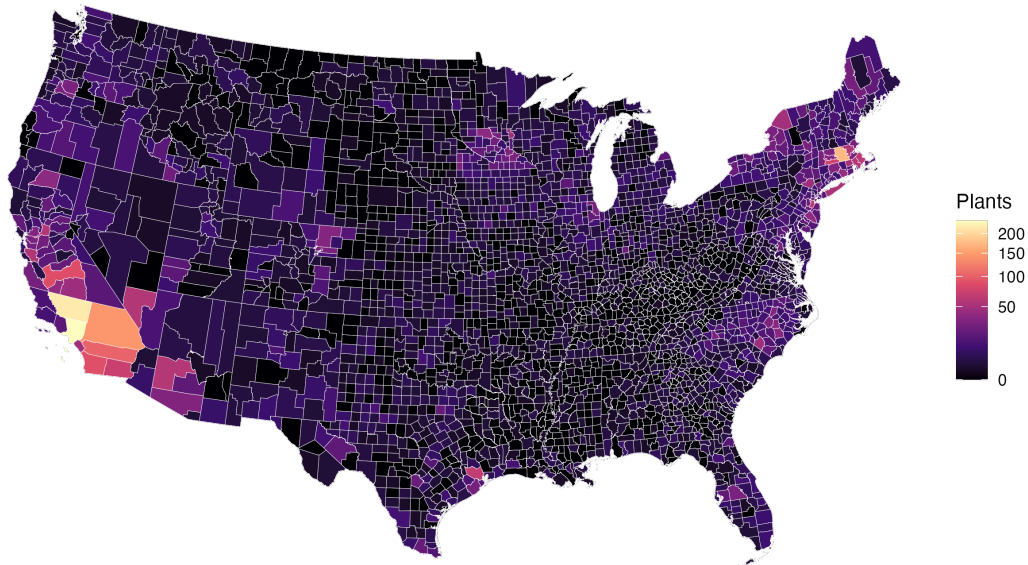


Figure 1: Plants per county

Q2: Plants within 25 km and 50 km. I buffered each plant by 25 km and 50 km and counted how many buffers intersect each county. I chose 25 km as a more conservative radius and 50 km to capture the larger zone where PM_{2.5} concentrations from taller stacks are elevated (Levy et al. 2009).

Q3: Mean PM_{2.5}. Used an exact extract to find each county's mean PM_{2.5} concentration. Comparisons listed below:

Counties within 25 km of any Plant

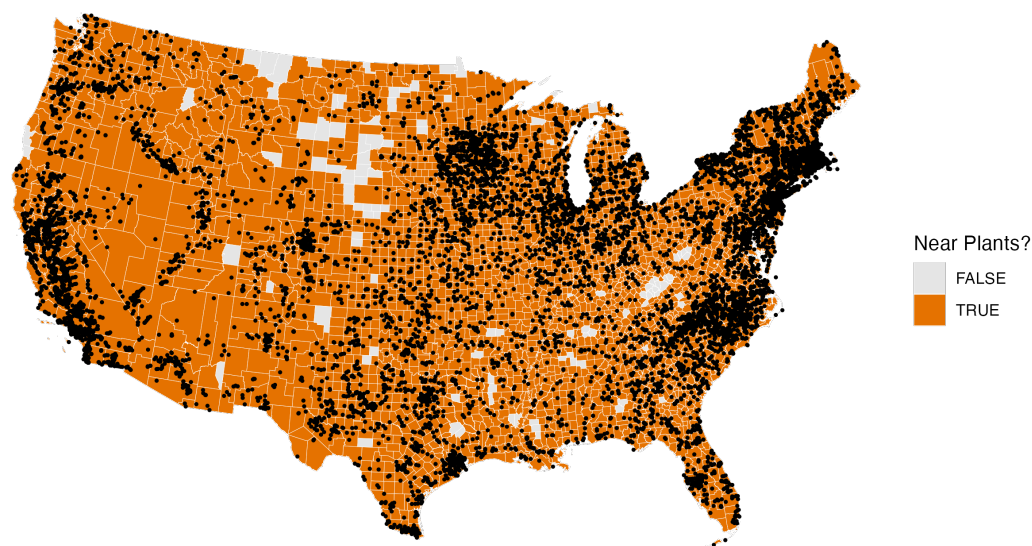


Figure 2: Counties within 25 km of any plant

Counties within 50 km of any plant

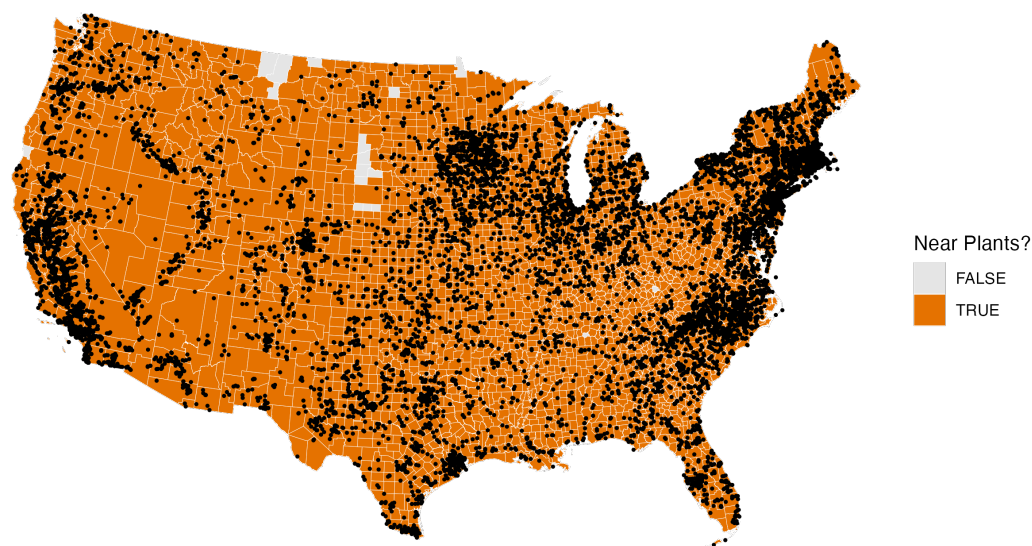


Figure 3: Counties within 50 km of any plant

Mean PM_{2.5} by county

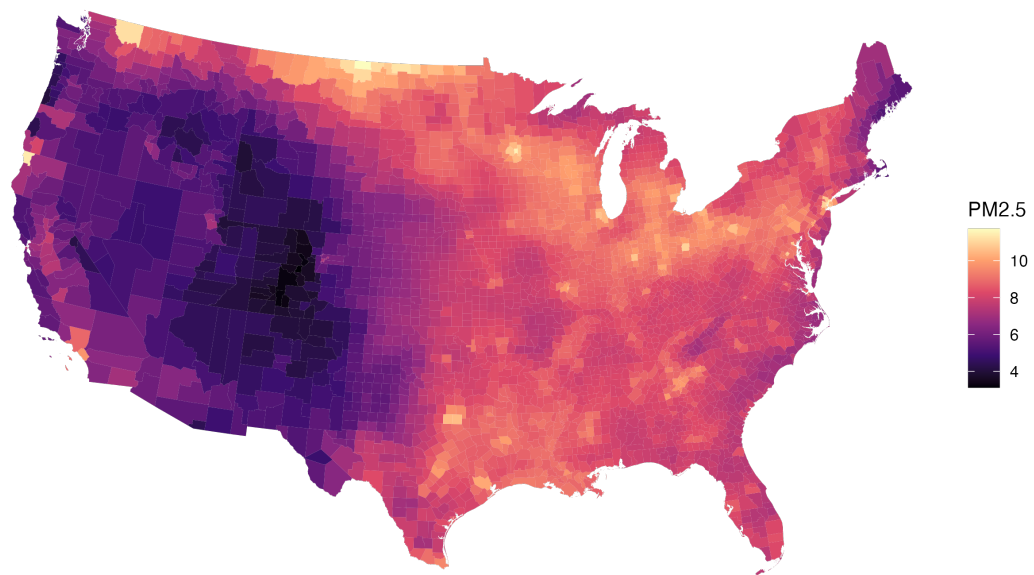


Figure 4: Mean PM_{2.5} by county

2 Comparison

Within-county definition:

Has Plants	Mean PM _{2.5} ($\mu\text{g}/\text{m}^3$)	Number of Counties
FALSE	7.99	974
TRUE	7.88	2,135

Nearby definition (25 km buffer):

Has Nearby Plants	Mean PM _{2.5} ($\mu\text{g}/\text{m}^3$)	Number of Counties
FALSE	7.86	107
TRUE	7.92	3,002

Nearby definition (50 km buffer):

Has Nearby Plants	Mean PM _{2.5} ($\mu\text{g}/\text{m}^3$)	Number of Counties
FALSE	8.14	18
TRUE	7.92	3,091

3 Discussion

Counties *without* plants have slightly higher mean PM_{2.5} than counties with plants (7.99 vs. 7.88). The nearby plants, however, flip the sign. Counties with plants within 25 km have slightly higher PM_{2.5} (7.92 vs. 7.86). Both differences are small ($\sim 0.1 \mu\text{g}/\text{m}^3$). At 50 km, the pattern is similar to the within-county result. The 18 counties with no nearby plants have a higher mean PM_{2.5} (8.14) than those with plants nearby (7.92). Given that there are only 18 counties without a nearby plant, however, this average is based on a very small number.

The picture changes slightly between the comparisons, but neither shows a strong raw relationship between power plants and PM_{2.5}. I'd like to emphasize that this is not a causal relationship. There are many features of counties that could impact PM_{2.5} levels outside of power plant presence. 69% of counties contain a power plant and 97% have one within 25 km, which also could limit the variation available to detect a difference.